

Netflix Success Prediction

Using IMDB Ratings and Catalog Metadata

DSA 210 – Introduction to Data Science

Elif Aksoy

Sabanci University

May 18, 2026

Abstract

This project investigates which measurable content attributes drive audience ratings on Netflix, and whether we can predict content success using only catalog metadata. Two Kaggle datasets were merged to produce a working dataset of 3,955 titles with genre, country, cast, director, and IMDB score information. After exploratory analysis and hypothesis testing, machine learning models were built for two tasks: predicting the exact IMDB score (regression) and classifying titles as successful or not ($\text{IMDB} \geq 7.0$). The best classifier, XGBoost with extended features, achieved 88.2% accuracy and an F1 score of 0.844. The key finding is that director and cast reputation are stronger predictors of success than genre or country of origin.

Contents

1	Motivation	2
2	Data Sources	2
2.1	Datasets	2
2.2	Data Collection and Merging	2
2.3	Data Enrichment	2
3	Data Analysis	3
3.1	Preprocessing	3
3.2	Feature Engineering	3
3.3	Exploratory Data Analysis	3
3.4	Hypothesis Testing	7
4	Machine Learning	8
4.1	Regression Results	8
4.2	Classification Results — Baseline Features	9
4.3	Classification Results — Extended Features	10
4.4	Feature Importance — Extended	11
5	Key Findings	11
6	Limitations and Future Work	12
6.1	Limitations	12
6.2	Future Work	13
7	AI Tool Usage Disclosure	13
8	References	13

1. Motivation

Netflix is the world’s largest subscription streaming platform, hosting tens of thousands of movies and TV shows from dozens of countries. Understanding what makes content succeed — and being able to predict it — has direct value for production decisions, acquisition strategies, and content recommendation systems.

This project investigates which measurable content attributes (genre, country of origin, release year, content type, cast, director) drive audience ratings on Netflix. Beyond descriptive analysis, the goal is to build predictive models that can classify a title as successful or not, and estimate its expected IMDB score, using catalog metadata and post-release signals such as audience vote counts, framing the analysis as a retrospective quality estimator rather than a pre-release predictor.

2. Data Sources

2.1 Datasets

Table 1: Summary of datasets used in this project.

Dataset	Source	Size	Key Features
Netflix Movies and TV Shows	Kaggle	8,807 titles	title, type, genre, country, release year, duration, content rating, director, cast
Netflix IMDB Scores	Kaggle	5,283 titles	title, type, IMDB score, vote count, runtime, age certification

2.2 Data Collection and Merging

Both datasets were downloaded from Kaggle. They were merged on the `title` column after normalizing to lowercase and stripping whitespace, using an inner join. This yielded **3,960 matched titles**. After dropping 5 rows with missing IMDB score or vote count, the working dataset contained **3,955 records**.

2.3 Data Enrichment

The enrichment strategy was to combine the Netflix catalog metadata with an independent IMDB ratings dataset. This produced a richer feature set than either dataset alone: the Netflix catalog has no rating or popularity data, while the IMDB dataset lacks genre, country, director, and cast fields. The merged dataset enables both EDA and machine learning across all dimensions simultaneously.

3. Data Analysis

3.1 Preprocessing

- `imdb_score` and `imdb_votes` were cast to numeric; 5 rows were dropped where either was missing.
- Missing `country` values ($\approx 36\%$) were filled with `'Unknown'` and excluded from country-level analyses.
- `duration` was parsed to extract an integer (minutes for movies, seasons for TV shows).
- Column aliases were cleaned (e.g., `type_x` $\rightarrow$ `type`).

3.2 Feature Engineering

Composite success score. A weighted combination of two normalized components was computed as a secondary target:

$$\text{composite_score} = 0.6 \times \tilde{s}_{\text{IMDB}} + 0.4 \times \tilde{s}_{\log\text{-votes}} \quad (1)$$

Both components were Min-Max scaled to $[0, 1]$. The log transform on vote count reduced the effect of outliers. The composite score captures both perceived quality and audience reach.

ML features (baseline): Content type (binary), release year, log vote count, runtime, label-encoded content rating, and binary indicators for the top 15 genres.

Extended ML features: Director and cast were LOO (Leave-One-Out) encoded as the mean IMDB score of all *other* titles by the same person, avoiding data leakage. The primary country was encoded as top-10 country dummies.

3.3 Exploratory Data Analysis

Score distributions. IMDB scores were approximately normally distributed, centered near 6.5. The composite score distribution was slightly right-skewed due to the popularity component.

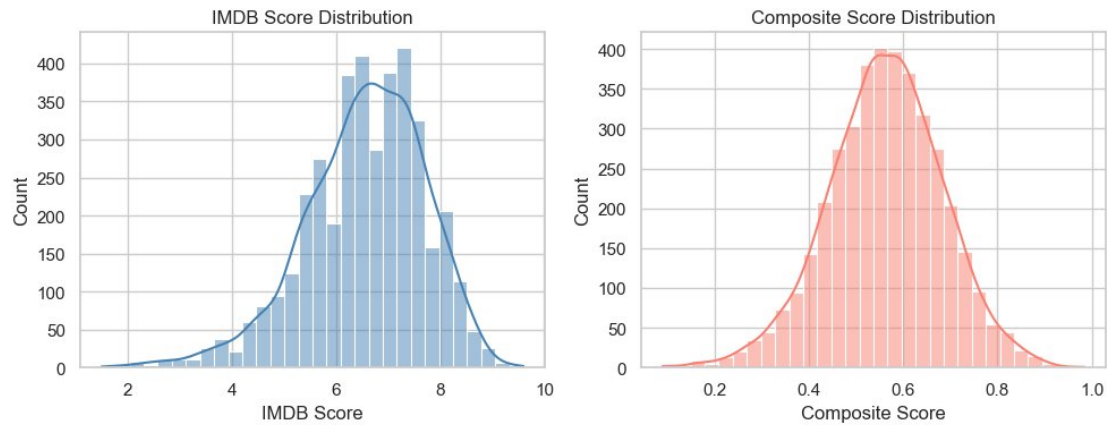


Figure 1: Distribution of IMDB scores (left) and composite scores (right) across 3,955 titles.

Content type. The dataset contained 2,461 movies and 1,494 TV shows. TV shows had a substantially higher mean IMDB score than movies (7.05 vs. 6.25).

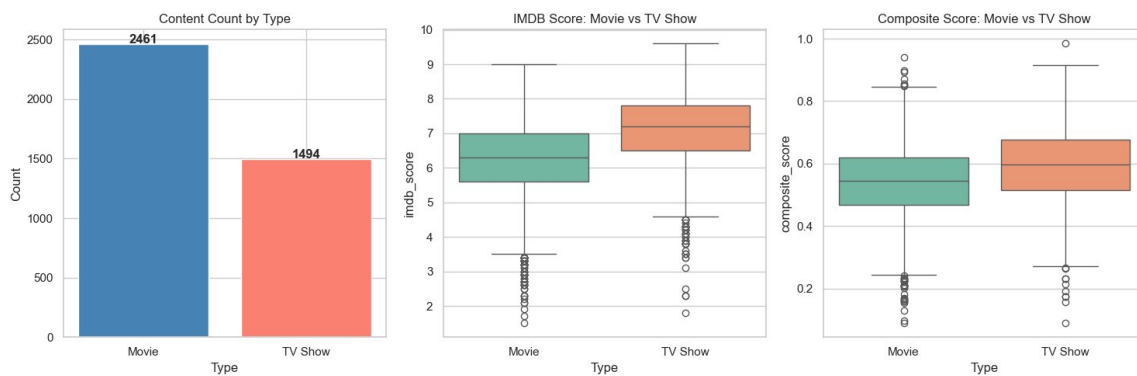


Figure 2: Content count by type (left), IMDB score (center), and composite score (right) for movies vs. TV shows.

Genre. Genre scores varied widely. Korean TV Shows, TV Thrillers, and TV Dramas scored highest. Horror Movies and Stand-Up Comedy were among the lowest-rated genres.

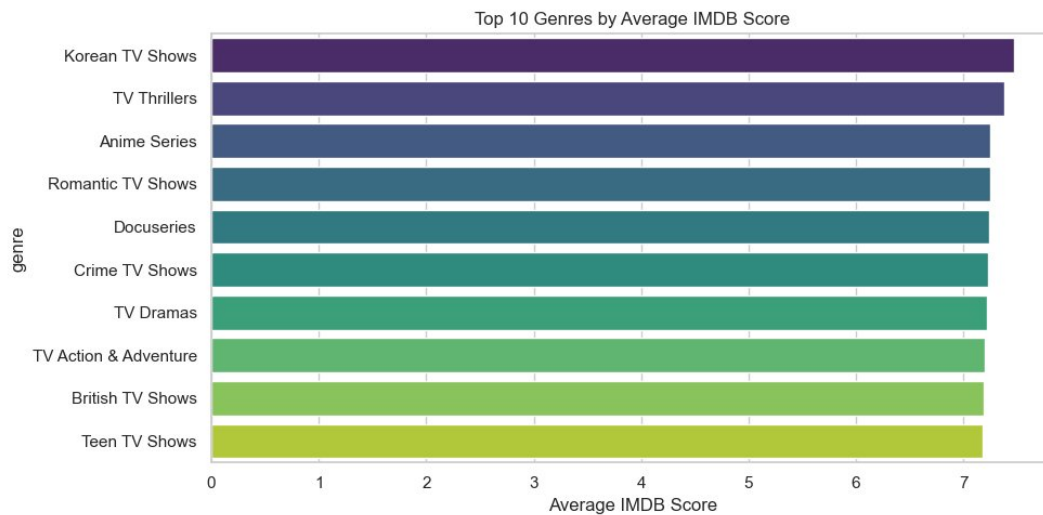


Figure 3: Top 10 genres by average IMDB score.

Release year. Content from across multiple decades was present, with recent years (2017–2021) dominating. No strong temporal trend was observed in ratings.

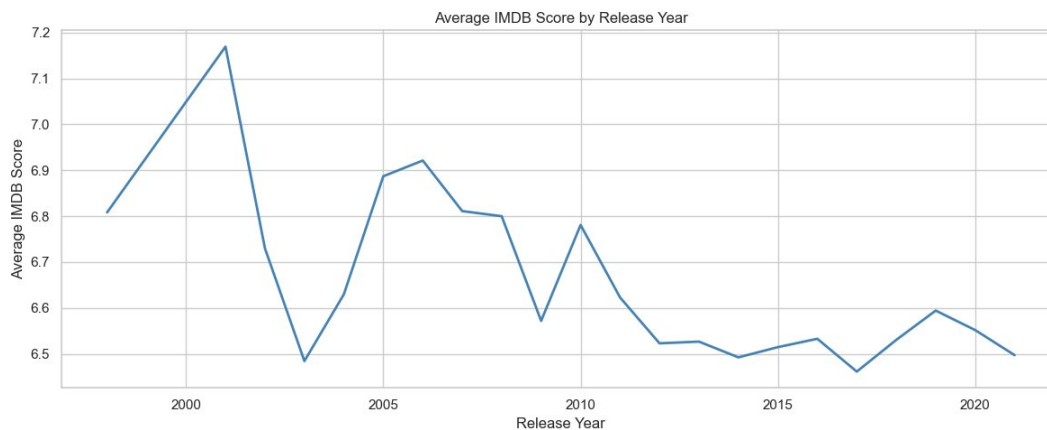


Figure 4: Average IMDB score by release year. No meaningful trend is visible.

Country. South Korean content (mean 7.24) and Japanese content (7.09) scored highest among countries with at least 20 titles. The United States contributed the most content ($n = 1,626$) but sat below the global average (6.57).

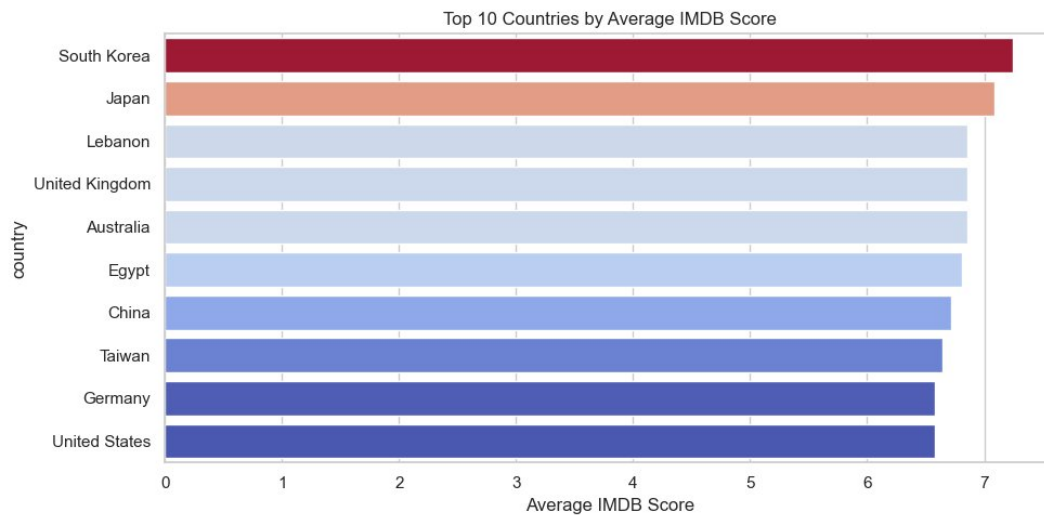


Figure 5: Top 10 countries by average IMDB score (minimum 20 titles per country).

Duration. Movie duration showed no strong relationship with IMDB score. For TV shows, longer-running series tended to score slightly higher, possibly reflecting survival bias.

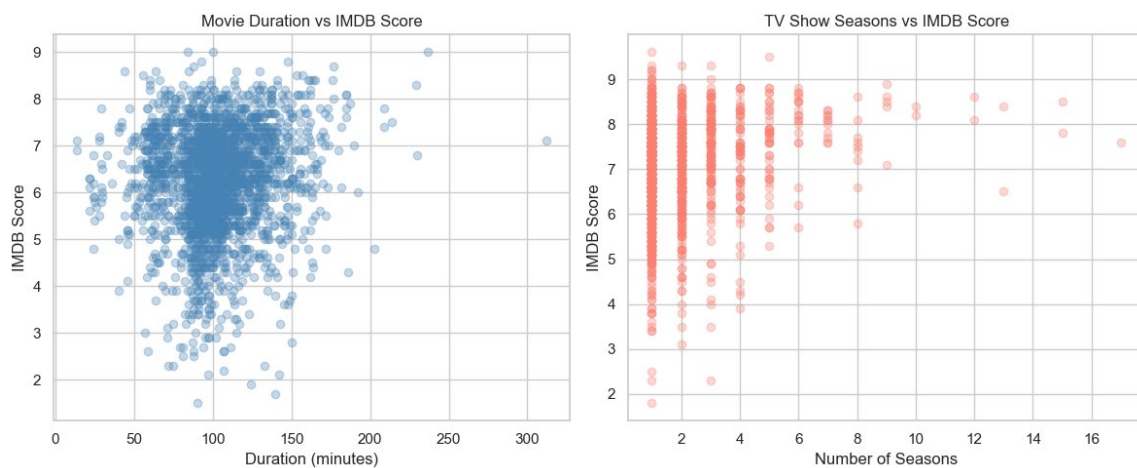


Figure 6: Movie duration vs. IMDB score (left) and TV show seasons vs. IMDB score (right).

Correlation heatmap. The strongest correlation was between IMDB score and composite score (0.84). Vote count had a moderate positive correlation with IMDB score (0.20).

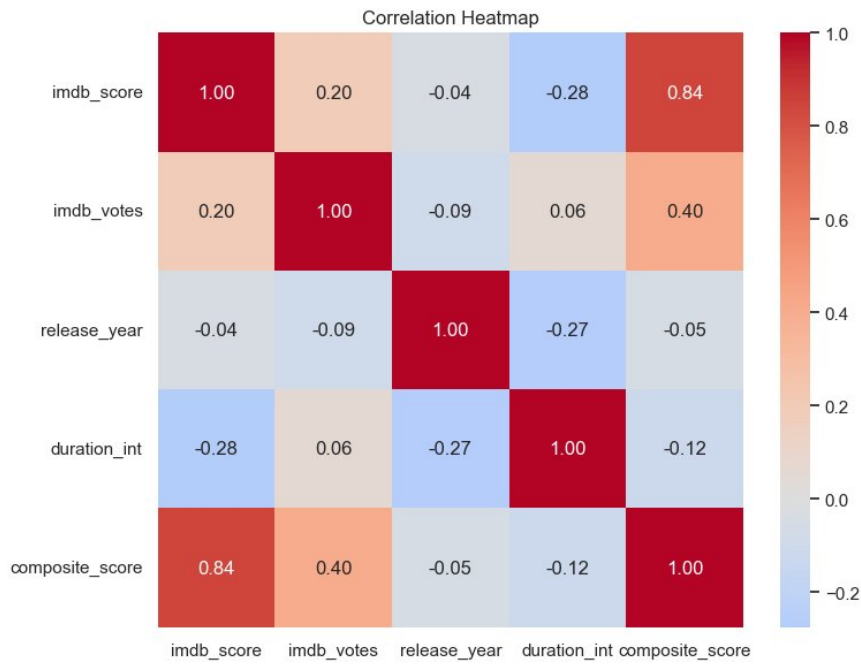


Figure 7: Pearson correlation heatmap of numerical features.

Popularity vs. quality. A modest positive correlation existed between vote count and IMDB score — more widely-seen titles tend to be rated somewhat higher — but popularity alone is not a reliable quality proxy.

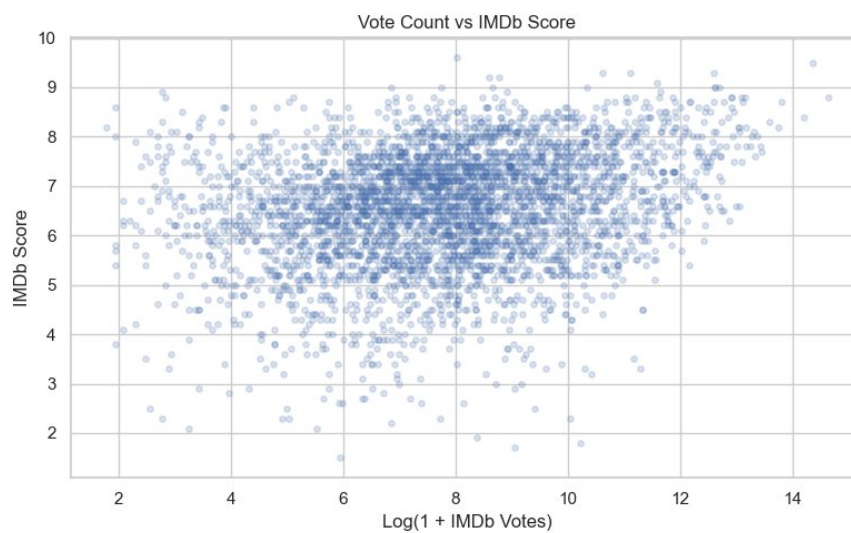


Figure 8: Log-transformed vote count vs. IMDB score. The positive trend is real but weak.

3.4 Hypothesis Testing

Five formal hypotheses were tested at significance level $\alpha = 0.05$:

Table 2: Summary of hypothesis tests.

#	Hypothesis	Test	Result
H1	Movies and TV shows have the same mean IMDB score	Two-sample t -test	Rejected ($p \approx 0$); TV shows score 0.80 pts higher on average
H2	All genres share the same mean IMDB score	One-way ANOVA	Rejected ($p \approx 0$, 34 genres); 296 of 561 genre pairs differ (Tukey HSD)
H3	Release year and IMDB score are uncorrelated	Pearson + Spearman	Not rejected in practice; Spearman $r = -0.018$, $p = 0.255$
H4	Vote count and IMDB score are uncorrelated	Pearson + Spearman	Rejected ; Pearson $r = 0.195$, Spearman $r = 0.239$, both $p \approx 0$
H5	All countries share the same mean IMDB score	One-way ANOVA	Rejected ($F = 8.18$, $p \approx 0$, 29 countries); South Korea and Japan lead

4. Machine Learning

Two tasks were modeled: regression (predict IMDB score) and binary classification (predict whether IMDB score ≥ 7.0 , the “successful” threshold). Four model families were evaluated for each task.

4.1 Regression Results

Table 3: Regression performance on the test set.

Model	RMSE	R^2
Linear Regression	1.004	0.302
Decision Tree	1.013	0.288
Random Forest	0.941	0.386
XGBoost	0.943	0.384

Random Forest and XGBoost performed comparably and both outperformed the linear baseline. $R^2 \approx 0.39$ indicates that the available metadata explains approximately 39% of score variance. The remaining variance is attributable to unobservable factors such as narrative quality, acting, and word-of-mouth.

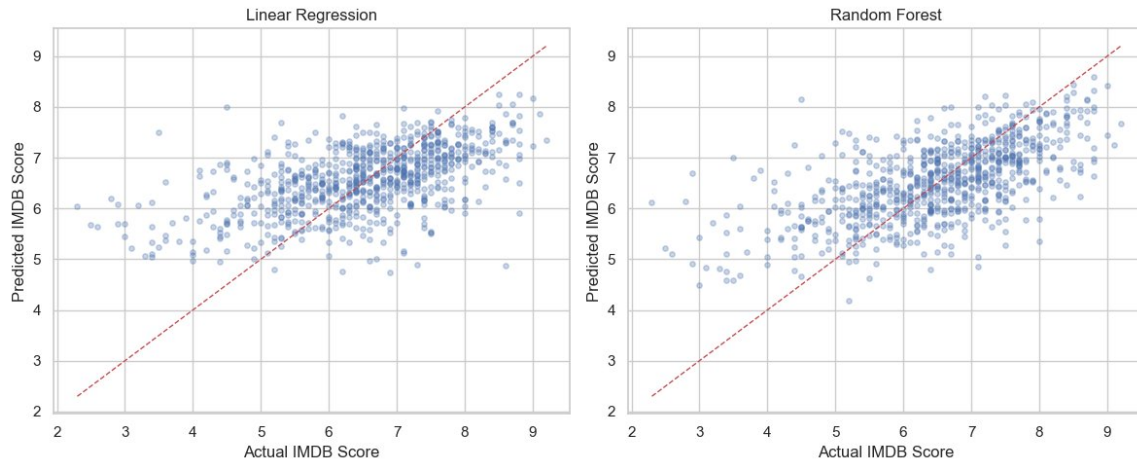


Figure 9: Predicted vs. actual IMDB scores for Linear Regression (left) and Random Forest (right). Points on the dashed line are perfect predictions.

4.2 Classification Results — Baseline Features

The binary label was defined as: success if IMDB score ≥ 7.0 (positive class $\approx 39.5\%$). F1 score was the primary metric due to mild class imbalance.

Table 4: Classification results with baseline features.

Model	Accuracy	F1
Logistic Regression	0.740	0.707
Decision Tree	0.743	0.633
Random Forest	0.770	0.697
XGBoost	0.798	0.732

XGBoost achieved the best performance. Logistic Regression with `class_weight='balanced'` achieved a higher F1 than Random Forest despite lower accuracy, confirming F1 as the more informative metric here.

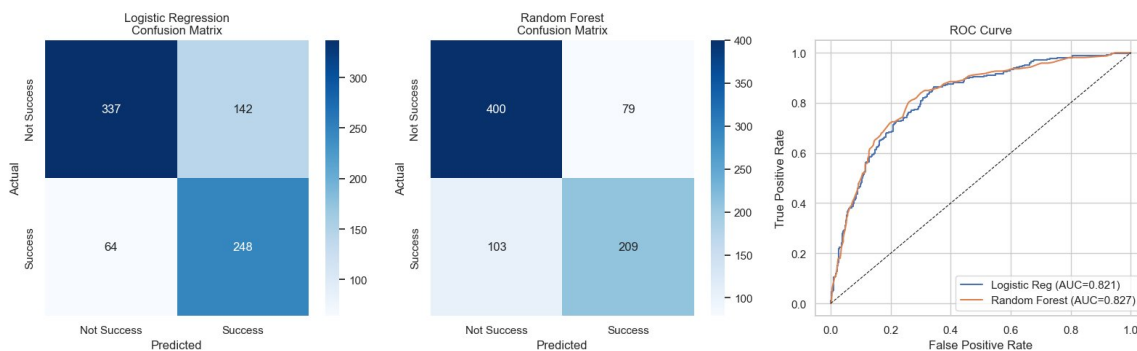


Figure 10: Confusion matrices for Logistic Regression (left) and Random Forest (center), and ROC curves for both models (right).

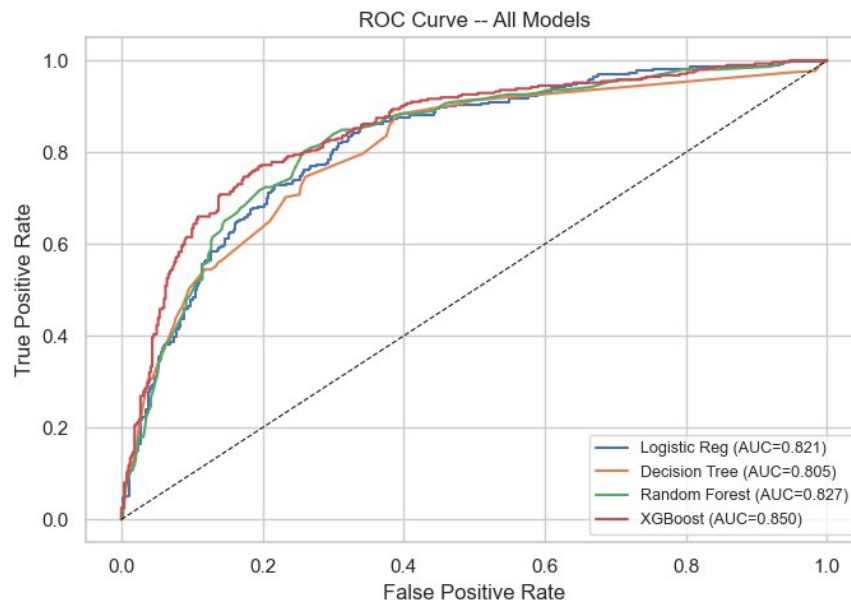


Figure 11: ROC curves for all four classifiers on the baseline feature set. XGBoost achieves the highest AUC of 0.850.

Feature importance — baseline. In the baseline Random Forest, `log_votes` was by far the most important feature, followed by `duration_int` and `release_year`. Genre indicators contributed only a small share of total importance.

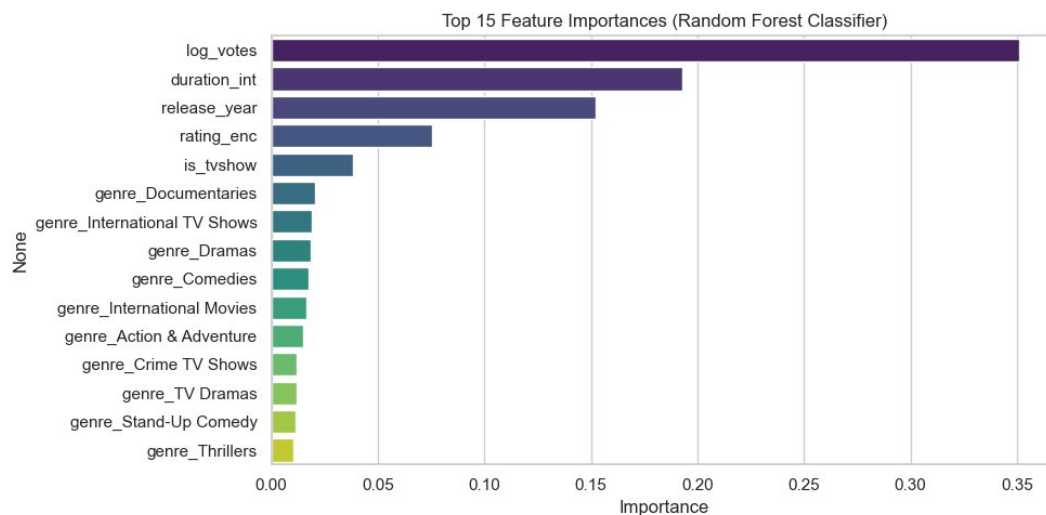


Figure 12: Top 15 feature importances from the baseline Random Forest classifier.

4.3 Classification Results — Extended Features

Adding director LOO encoding, cast LOO encoding, and country dummy variables produced substantial improvements across all models.

Table 5: Classification results with extended features (+ director, cast, country).

Model	Accuracy	F1
Logistic Regression	0.765	0.703
Decision Tree	0.837	0.784
Random Forest	0.867	0.821
XGBoost	0.882	0.844

XGBoost F1 improved from 0.732 to 0.844 (+0.112). Random Forest F1 improved from 0.697 to 0.821 (+0.124). The ROC AUC for XGBoost on the extended feature set reached approximately 0.93.

4.4 Feature Importance — Extended

In the extended Random Forest model, `director_loo` and `cast_loo` ranked among the top features alongside `log_votes`. This confirms that production team reputation is at least as informative as audience popularity in predicting content success.

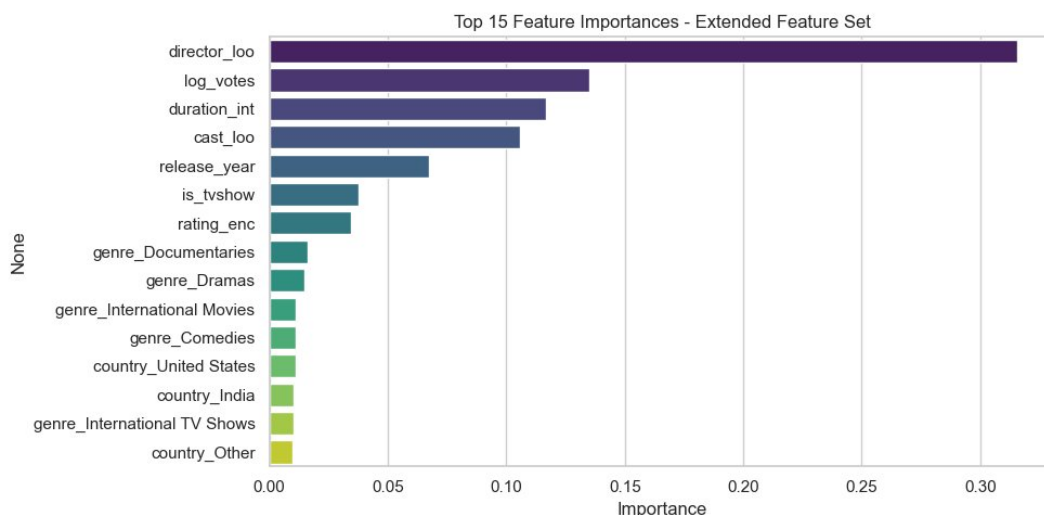


Figure 13: Top 15 feature importances from the extended Random Forest classifier. Director LOO encoding is the single most important feature.

5. Key Findings

- TV shows outperform movies.** TV shows have significantly higher IMDB scores (mean 7.05 vs. 6.25, $p \approx 0$). This likely reflects that TV audiences self-select into series they expect to enjoy, whereas movies attract broader, more casual viewers.
- Genre is a strong predictor of quality.** ANOVA and Tukey HSD confirmed significant rating differences across 34 genres. Korean TV Shows consistently out-

perform other genres. Horror Movies are rated significantly lower than most other genres.

3. **Release year has no meaningful effect.** Although a tiny negative correlation is statistically significant at $n = 3,955$, it is practically negligible (Spearman $r = -0.018$, $p = 0.255$).
4. **Country of origin matters.** South Korea (7.24) and Japan (7.09) produce the highest-rated Netflix content per title. Despite contributing the most titles, the United States (6.57) sits below average — suggesting that volume and quality are inversely correlated at country level.
5. **Popularity correlates with quality, but weakly.** Vote count and IMDB score are positively correlated (Spearman $r = 0.239$). Popular titles tend to score higher, but popularity alone is insufficient for prediction.
6. **Director and cast reputation are the most powerful predictors.** The LOO-encoded director and cast features produced the largest gain in classification performance (+0.112 F1 for XGBoost). A title is most accurately predicted by *who made it*, not just what genre or country it belongs to.
7. **Metadata explains ~39% of score variance.** The best regression model (Random Forest, $R^2 = 0.386$) leaves most of the variance unexplained. IMDB scores are ultimately driven by subjective viewer experience, which catalog metadata cannot fully capture.

6. Limitations and Future Work

6.1 Limitations

- **Merge coverage:** The inner join retained 3,955 of 8,807 Netflix titles (~45%). Titles with slightly different spellings were lost, potentially introducing selection bias.
- **Data leakage caveat:** `log_votes` is a post-release signal. This makes the model a retrospective quality estimator, not a true pre-release predictor. The LOO director/cast features are also retrospective.
- **Missing metadata:** 36% of titles have unknown country; 36% lack director information. These were imputed with global means or 'Unknown'.
- **Static snapshot:** Both datasets represent a fixed point in time. Netflix catalog and ratings change continuously.

- **Binary success threshold:** Defining success as $\text{IMDB} \geq 7.0$ is arbitrary. Different thresholds yield different class balances and model behaviors.

6.2 Future Work

- **Pre-release prediction:** Remove post-release features and use pre-production signals such as budget and marketing spend.
- **NLP on descriptions:** Apply TF-IDF or sentence embeddings to plot synopses to extract thematic signals correlated with quality.
- **Temporal modeling:** Capture how a title's rating evolves after release.
- **Recommendation system:** Extend the framework into a personalized content recommendation engine.
- **Rotten Tomatoes / Metacritic integration:** Add critic scores to model the gap between critical and audience reception.

7. AI Tool Usage Disclosure

As required by the academic integrity policy, AI tools were used during this project. **Claude (Anthropic)** assisted with parts of the code implementation, report writing, and a final consistency check. In all cases, the core ideas, analytical decisions, implementations and interpretations were the author's own — Claude was used strictly as a support tool.

8. References

1. Netflix Movies and TV Shows. Kaggle.
<https://www.kaggle.com/datasets/shivamb/netflix-shows>
2. Netflix TV Shows and Movies (IMDB Scores). Kaggle.
<https://www.kaggle.com/datasets/victorsoeiro/netflix-tv-shows-and-movies>
3. Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
4. Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of KDD '16*.